Subject: Reinforcement Learning

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Assignment 1

SemEval 2027 Task 9, Subtask 2:
Image Framing in Multilingual News

Group #12

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Abstract

This report looks at SemEval 2027 Task 9, Subtask 2, where a system assigns zero or more frame labels to a news image. The first supplied training chunk has 400 English image–article pairs. We checked every image-to-article join and examined the labels and file quality before training. We then compared a small set of handcrafted image features with frozen ResNet-18 features, both with and without article text. On a shared 280/60/60 fit/calibration/evaluation split, local micro-F1 was 0.272 and 0.283 for the classical models and 0.309 and 0.308 for the frozen models. These are internal results, not organizer scores. The supplied Subtask 1 articles do not give us gold text-frame labels for the 400 Subtask 2 articles, so we could not run a valid 2b experiment on this chunk. The official scorer and later data releases still need to be checked.

1 Introduction and Task Definition

The organizer's Subtask 2 [1] takes a news image and its associated article and predicts the *image's* frames from a shared taxonomy of 14 generic categories. This is multilabel classification: an image may express several categories or none. In 2a, article text is context but gold article-frame labels are unavailable. In 2b, those gold text-frame labels are supplied as additional context; they are never image targets. The task spans Bulgarian, English, European Portuguese and Russian, although the attached Subtask 2 training release contains only English.

The task is difficult because an abstract frame may not correspond to one recognizable object. The article can clarify an image or point toward a different interpretation. Image crops, embedded text, source templates and repeated photographs may also give a model shortcuts. Annotation imbalance matters too: one frame is entirely absent from this first release. The organizer specifies multilabel micro-F1 as the primary metric. Our treatment of empty-label rows remains provisional until we can check the official scorer (Section 7).

2 Related Work

Card et al. [4] built the Media Frames Corpus from policy-news text. It gives useful background for the frame categories, although labeling spans of text is different from labeling an image. Akyürek et al. [2] study multilabel framing across languages using headlines. This is relevant to the wider SemEval task, but our supplied chunk is English only and its target is visual. Piskorski et al. [6] also study multilingual news framing in a SemEval setting. Their framing component is text based, so its scores are not a direct comparison with ours.

Tourni et al. [7] combine lead images with news text in a study of gun-violence coverage. Their finding that image relevance affects the value of multimodal information gave us a reason to keep an image-only comparison alongside 2a. Their target and single-topic dataset differ from this image-frame task. Arora et al. [3] study framing across news text and images more broadly. That work reinforces the need to examine both modalities without treating their labels as interchangeable. For the visual baseline, we use the residual network introduced by He et al. [5], with its pretrained weights frozen. Object-recognition features are a starting point, not a claim that ResNet-18 understands abstract frames. No organizer baseline paper was included with this first release; we would review it before a later shared-task submission.

3 Dataset

The supplied `data.zip`, reconstructed from three uploaded parts, is 84,957,589 bytes with SHA-256 `ad18b9e877abb2f6a469bc7df0e855bebc684ead6a81d0e332765bff8e071fa9`. The ZIP integrity check

passes. Its Subtask 2 portion calls itself *training data, chunk 1.* It has 400 English images, 400 UTF-8 articles, 400 optional HTML views and 400 image-annotation rows. We join `EN_doc_N_image.jpg` to `EN_M_TRAIN_doc_N.txt` only after checking both files and all document IDs 1–400. The exact 14-label vocabulary agrees across the archive's two Subtask 1 label files and validates every image annotation. The Subtask 1 articles are a separate corpus, so their annotations cannot be used as gold 2b context for these 400 images.

Audit item	Count
English labeled image–article pairs	400
Missing or corrupt images/articles	0
Images with no frame	72
Positive image-label assignments	659
Observed labels of 14	13
Exact duplicate images/articles	0/0
Identical difference-hash image candidates	0

Table 1: Audit of the attached first English Subtask 2 training chunk. No official development or test images are in this archive.

Of the 400 images, 165 have one label, 90 have two, 42 have three and 31 have four or more; four of the latter have 13 labels and deserve annotation review. Article length ranges from 113 to 8,366 whitespace-separated words (median 514.5). The absence of exact duplicates does not exclude common news templates or related photographs. Table 2 shows substantial label imbalance; *Legality, constitutionality and jurisprudence* has no positive image.

Frame	Count	Frame	Count
Quality of life	113	Political	85
Economic	78	Cultural identity	78
Health and safety	72	Public opinion	45
Capacity and resources	44	Security and defense	38
Crime and punishment	34	Fairness and equality	23
Policy prescription and evaluation	20	Morality	17
External regulation and reputation	12	Legality, constitutionality, jurisprudence	0

Table 2: Image-label frequencies in the 400 training pairs. Display names are shortened; code uses the exact underscore-separated strings in the archive.

4 Exploratory Data Analysis

A separate single PDF, `artifacts/release1-eda.pdf`, contains labeled plots of label frequency and co-occurrence, aspect ratio, brightness, saturation, edge density, article length, standardized frame-feature differences and language coverage. These five feature groups (geometry, light, color, visual edge structure and article length) were selected to probe possible visual cues and shortcuts; label balance and file quality were audited separately. Associations compare the positive and negative feature means in units of overall feature standard deviation. They are descriptive, especially for rare frames.

Observation. 167 of 400 images are exactly 1200 by 630 pixels, and all supplied Subtask 2 examples are English. *Interpretation:* a repeated display size may reflect publishing templates, while the language restriction prevents a cross-language conclusion. *Modeling implication:* geometry is useful for auditing, but source/template effects need explicit grouping or metadata in later releases; these results must not be extrapolated to other languages.

Observation. Images tagged *Quality of life* have edge density 0.54 standard deviations above the remaining images and *Political* images have mean brightness 0.42 standard deviations below the remainder. *Interpretation:* these differences might reflect topic, source or cropping rather than a causal frame cue. *Modeling implication:* retain edge and brightness statistics in the interpretable baseline, inspect their coefficients and test their stability when later data are available.

Observation. Article length has median 514.5 words and a maximum of 8,366. *Interpretation:* one very long article can affect lexical features, but length alone does not prove framing. *Modeling implication:* use sublinear TF-IDF fitted on the fit partition only and compare with an image-only model to expose text-driven shortcuts.

5 Feature Engineering

The classical image vector contains 59 values: log width, height, area and aspect ratio; mean and dispersion of brightness, saturation and RGB channels; spatial brightness; edge density and strength; and grayscale/RGB histograms. Images are EXIF-oriented, converted to RGB and reduced to at most 256 pixels per side for these statistics. Article context is word-and-bigram TF-IDF (up to 20,000 features, sublinear term frequency), with the vocabulary fit only on the 280 fit examples. The frozen visual representation is the 512-dimensional penultimate output of torchvision ResNet-18 with `IMAGENET1K_V1` weights [5]; its standard 224-pixel preprocessing is used. The encoder stays frozen. OCR and visual–text similarity are not included in this first baseline.

6 Baseline Models

For each of the 14 image frames, we fit a separate class-weighted regularized logistic-regression head. A frame with no positive fit examples receives a recorded constant head, rather than a spurious trained classifier. Each representation is evaluated with image features alone and image features plus article TF-IDF (2a). Gold text-frame input for 2b is intentionally omitted because the release does not provide it for these articles. The two representation families and two input settings share a split and calibration procedure.

Coefficients in the classical image-only heads provide a limited diagnostic: *Quality of life* has a positive standardized edge-density coefficient (0.55), consistent with the EDA association. Several grayscale histogram bins have larger absolute coefficients, but the features are strongly correlated. Their coefficients cannot support a simple causal interpretation. The coefficient pattern is a hypothesis for future data, not proof that edges convey this frame.

7 Evaluation Setup

With seed 42, a deterministic search over grouped splits favors frame balance and produces 280 fit, 60 threshold-calibration and 60 held-out evaluation images. This split search uses the training

labels to balance the partitions, including the eventual evaluation rows. The group key is the article/document ID; no exact image/article duplicates or identical difference hashes were found, but source templates could still span groups. The same split CSV is used for all four variants. It contains 463, 105 and 91 positive labels in fit, calibration and evaluation respectively; the evaluation also contains ten unframed images. A single probability threshold per model is selected on calibration micro-F1 from 0.10 to 0.90 in increments of 0.05. No held-out labels enter scaling, TF-IDF fitting, model fitting or threshold selection.

For totals over all image/frame decisions, micro-F1 is

$$F_{1,\text{micro}} = \frac{2TP}{2TP + FP + FN}. \tag{1}$$

It pools the 14 frames, so common frames influence it more than rare ones. Macro-F1 averages per-frame F1 and exposes unsupported categories. The organizer's task description mentions a `None` convention ambiguously for Subtask 2. Our provisional local score uses `-none-mode exclude`: no extra positive class is created, but predicting frames for an unframed image still adds false positives. An official scorer was not supplied, so these are internal scores only.

8 Results

Table 3 reports one held-out evaluation on the supplied English chunk. The frozen image-only representation has the highest local micro-F1 at 0.309; adding article text changes it by only 0.001. The classical 2a run improves by about 0.011 over its image-only version. These small differences from one split do not establish a dependable model ranking. All four systems predict frames for each of the ten unframed evaluation images. The frozen thresholds reach the search's lower bound (0.10), indicating calibration sensitivity and substantial false positives.

Representation	Input	Threshold	TP	FP	FN	Micro-F1	Macro-F1
Classical	Image only	0.45	50	227	41	0.272	0.218
Classical	Image + article (2a)	0.35	51	219	40	0.283	0.219
Frozen ResNet-18	Image only	0.10	46	161	45	0.309	0.224
Frozen ResNet-18	Image + article (2a)	0.10	46	162	45	0.308	0.223

Table 3: Provisional local scores on the common 60-image held-out split. These are not organizer scores.

Per-frame F1 for the frozen 2a run ranges from 0 for several rare categories to 0.50 for *Political*. The absent legality category has zero support and a constant head. The apparent micro-F1 advantage over macro-F1 reflects skew and weak rare-label behavior, not broad reliability.

9 Error Analysis

The saved per-image predictions support concrete checks. Frozen 2a exactly predicts *Economic* and *Quality of life* for `EN_doc_25_image.jpg` and *Quality of life* plus *Cultural identity* for `EN_doc_53_image.jpg`. It misses every gold frame on `EN_doc_104_image.jpg` (three frames) and predicts four frames on the unframed `EN_doc_20_image.jpg`. Classical 2a is more prone to overprediction: for `EN_doc_16_image.jpg`, with two gold frames, it emits seven. These examples show both omission and spurious co-occurrence. They do not by themselves identify whether the visual features, the article context

or annotation ambiguity caused an error; inspection of the raw images and articles is necessary for that attribution.

10 Discussion

The supplied ZIP, importer, quality audit, EDA, four model variants, TSV response writer and local scorer now run end to end. The 400-example release is much more informative than the 20-example public trial, but still only the first English training chunk. Four images with 13 labels, 72 unframed images, an absent taxonomy category, skewed labels and possible source templates constrain interpretation. Article TF-IDF produces little change relative to image-only features on this split; that does not show that articles are irrelevant to image framing. The 2b question remains open because the required paired gold text-frame context is absent. The official scorer, later language data and final submission contract have not been verified. The saved heads are fit on the 280-example fit subset and are research baselines, not final refits for organizer submission.

11 Next Steps

Obtain later training chunks, paired 2b text frames when released, the official starter kit/scorer and any organizer baseline paper. Verify empty-frame scoring and submission syntax, then rerun grouped evaluation with source/template and near-duplicate controls. Review the four 13-label items and representative false positives/negatives. Consider OCR where embedded image text is common. After model selection, refit on all permitted labeled training data and validate the response file with organizer tools. The supplied Assignment 1 guidelines do not specify a required filename, upload portal, or due date; these should be confirmed with the instructor.

Reproducibility

Use Python 3.11–3.13 and install `requirements.txt`. First run `python -m framing prepare-release` on `data/release1/data.zip`. Then run the `audit`, `eda` and `train` commands with the generated manifest and label list. Use seed 42, the `exclude` setting for empty-label scoring and separate output folders for classical and frozen models. Install `requirements-pretrained.txt` before the frozen run. The submission README gives every command and path in full. Do not enable 2b for this release. The archive, audit, EDA, split, predictions, metrics and package versions are tracked with the code. These runs used Python 3.12.14, scikit-learn 1.9.1, torch 2.5.1+cpu and torchvision 0.20.1+cpu on a CPU-only runner.

References

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